

TESTING AND VERIFICATION REPORT

Developer: [Left Blank] | OptiCrop Project Documentation

Repository: <https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine>

1. Model Accuracy Verification

To verify system reliability, we compared four classification models on the crop dataset:

Algorithm	Training Accuracy	Test Accuracy	Weighted F1-Score
Random Forest Classifier	100.0%	100.0%	1.00
Decision Tree Classifier	100.0%	98.1%	0.98
K-Nearest Neighbours	100.0%	97.3%	0.97
Logistic Regression	96.2%	95.1%	0.95

Opticrop - Testing & Performance

Model Accuracy (validation set)

Random Forest Classifier : 100.00 %

Decision Tree Classifier : 98.18 %

K-Nearest Neighbours : 97.50 %

Logistic Regression : 95.68 %

2. Web Interface and System Checks

Yield Regressor

- Forms: Verified that submitting empty fields triggers error responses.

Random Forest Regressor : R2 = 0.9249

- Latency: Tested local server predictions, returning HTML templates in under 50ms.

- Data Persistence: Verified that predictions are successfully written to the CSV logs.

Automated Unit Tests (pytest / unittest)

test_homepage_loads PASSED

test_predict_endpoint PASSED

test_csv_logging PASSED

test_bounds_validation PASSED

=> 4 / 4 tests passed